Year 13 Maths - Sequences and series

Similar questions - Recurrence relations and models

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A sequence is defined by $u_1 = 3$ and $u_{n+1} = 1 - \dfrac{1}{u_n}$ for $n \geqslant 1$.

(a) Find u_2, u_3 and u_4. **[3]**

(b) Describe the behaviour of the sequence. **[1]**

(c) Find $\sum_{n=1}^{3} u_n$ and hence $\sum_{n=1}^{100} u_n$. **[4]**

2 Two savings plans both start with a deposit of £200 in month 1. Under plan A, each month's deposit is £25 more than the previous month's. Under plan B, each month's deposit is 6% more than the previous month's.

(a) Find the deposit in month 12 under each plan. **[3]**

(b) Show that, in the month n in which the deposit under B first exceeds that under A, n satisfies $1.06^{n-1} > 1 + 0.125(n-1)$. **[3]**

(c) Find the value of n, by calculation. **[2]**

(d) Give one reason why each plan may not be realistic over a long period. **[2]**

Answers

Question 1

(a) $u_2 = 1 - \frac{1}{3} = \frac{2}{3}$. $u_3 = 1 - \frac{3}{2} = -\frac{1}{2}$. $u_4 = 1 + 2 = 3$

(b) $u_4 = u_1$, so the sequence repeats $3, \frac{2}{3}, -\frac{1}{2}$ with period 3 (a periodic sequence)

(c) $3 + \frac{2}{3} - \frac{1}{2} = \frac{19}{6}$. $100 = 33 \times 3 + 1$, so the sum is $33 \times \frac{19}{6} + u_1 = 104.5 + 3 = 107.5$

Question 2

(a) A: $200 + 11 \times 25 = 475$. B: $200 \times 1.06^{11} = £379.66$

(b) B exceeds A when $200(1.06)^{n-1} > 200 + 25(n-1)$. Divide by 200: $1.06^{n-1} > 1 + 0.125(n-1)$

(c) At $n = 25$: $B = 809.79$ and $A = 800.00$; at $n = 24$: $B = 763.95$ and $A = 775.00$. So $n = 25$

(d) A: a deposit rising by a fixed £25 every month for years would become unaffordable or exceed income. B: a 6% monthly increase compounds very rapidly and would also become unaffordable (or the person's income will not keep pace)